TAC ECONOMICS

MACROECONOMETRIE

2026-2027

General informations

- Virginie Gautier
- Economist – data scientist chez TAC ECONOMICS
- PhD in Economics (Université de Rennes/CREM/EDGE)
- virginie.gautier@taceconomics.com
- GitHub: https://github.com/vgautier1/cours_M2_IEF
- **Assignments due by October 4**

Course structure

- 8 * 3.5-hour sessions co-taught with Florian Pothin
- 4 macroeconomic concepts, 7 hours per concept
- Exam:
  - Project to submit at the end of each concept (report + data + code/script)
  - 2 out of 4 projects graded, or average of the 4 projects (to be confirmed)
- Report content:
  - 1–3 pages on theory + introduction of the research question + stylized facts + outline
  - 2–5 pages on the econometric application (explanation of methodology and results + tests)
  - 1–2 pages of conclusion (link between theory and econometric results)

Be careful with the use of generative AI

PHILLIPS CURVE – REVIEW

Overview

- Illustrates a negative empirical relationship between unemployment rate and inflation rate
- Central concept in macroeconomics, especially for economic policymaking
- Challenged during the 1970s
- Modern Phillips curve and limitations

I- Origins (1950s-1960s)

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- A.W. Phillips (1958) studied the macroeconomic situation of the UK between 1861 and 1913 and found an inverse relationship between the unemployment rate and the rate of change of nominal wages.
- When unemployment is high, nominal wages increase slowly because workers are not in a position to bargain due to the large available labor force. The opposite effect is observed when unemployment is low: workers are in a strong bargaining position due to scarce available labor and can negotiate nominal wage increases.

⇒ ***Negative relationship between unemployment and wage inflation***

- Neoclassicals: wages are linked to the productivity of labor

⇒ ***The Phillips curve contradicts neoclassical theory***

- Keynesians: active economic policies, such as stimulating aggregate demand, can influence unemployment and inflation.

⇒ ***The Phillips curve supports Keynesian thought***

I- Origins (1950s-1960s)

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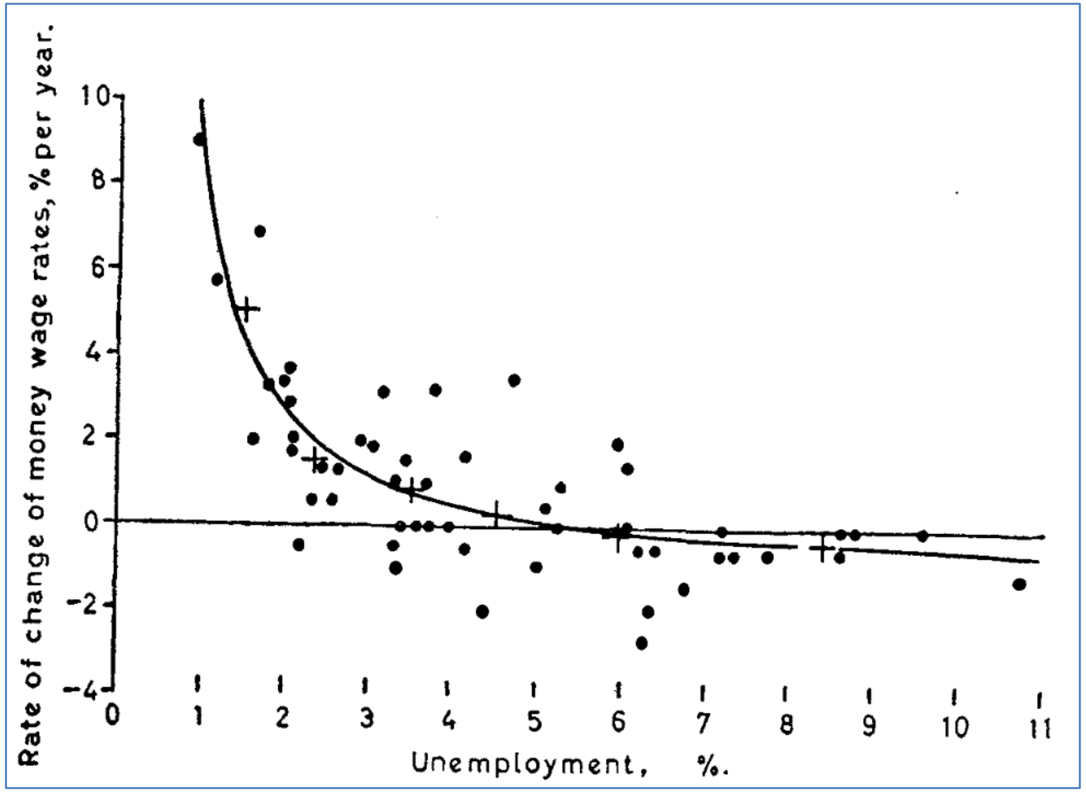
- The original Phillips curve describes a **negative** and **non-linear** relationship between the unemployment rate and the growth rate of nominal wages.
- **Empirical** relationship, as it is based on observations.

$$\frac{\Delta W}{W} = \beta U \qquad (1)$$

W nominal wages et U unemployement rate.



Source: Phillips (1958).

I- Origins

- Theorization of the Phillips curve by R. Solow (Keynesian) and P. Samuelson (Neoclassical) when focusing on the US.

=> They extended Phillips' relationship to inflation because wage increases are passed on to prices (via production costs) and to demand (gains in purchasing power).

- **Integration into the Keynesian model:**
  - <u>Unemployment</u>: labor market disequilibrium caused by insufficient aggregate demand.
  - <u>Phillips Trade-off:</u> Stimulating aggregate demand reduces unemployment (via higher output) but accelerates inflation due to wage and price pressures.

⇒ ***Policymakers must choose an 'optimal' equilibrium point allowing moderate inflation and unemployment according to their preferences, or accept higher inflation to boost demand/reduce unemployment.***

II- Challenging the Theory (1970s) – Phillips Curve Horizon

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- Critique by M. Friedman and E. Phelps: the **inverse relationship between unemployment and inflation disappears in the long run**. Their argument is based on the fact that economic agents (consumers, firms) adapt their inflation expectations based on past conditions, a concept known as **adaptive expectations**.
- Gives rise to the expectations-augmented Phillips curve:

$$\pi_t = \pi_t^e - \alpha\left(U_t - U_t^N\right) \qquad (2)$$

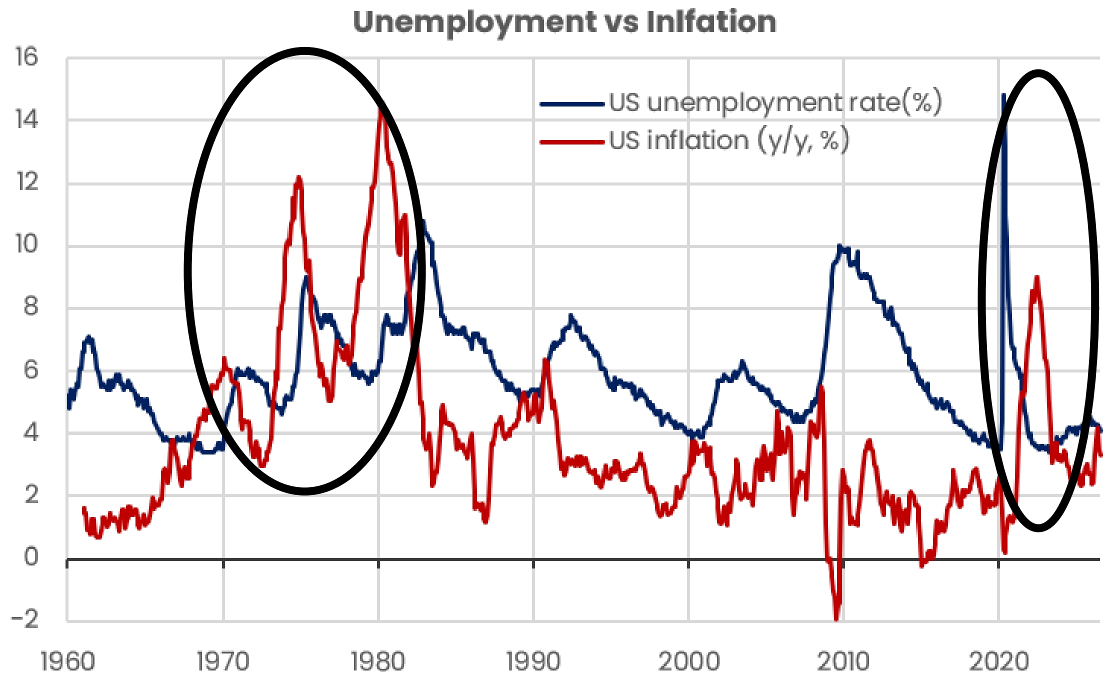
π_t^e *expected inflation and* U_t^N *the natural rate of unemployment (NAIRU).*

- Friedman and Phelps introduced the concept of NAIRU, representing a level of unemployment compatible with stable inflation. At this level, there is no longer a trade-off between unemployment and inflation. Any attempt to reduce unemployment below this level through demand stimulus policies simply leads to an acceleration of inflation.
- ***The trade-off between unemployment and inflation highlighted by the traditional Phillips curve is only valid in the short run. In the long run, expansionary policies aimed at maintaining artificially low unemployment only increase inflation without any lasting effect on unemployment (vertical long-run Phillips curve).***

II- Challenging the Theory (1970s) – Accounting for Supply Shocks

- The stagflation (coexistence of high unemployment and high inflation) of the 1970s challenged the foundations of the traditional Phillips curve.
- 1970s: Major supply shocks, notably oil shocks, increased production costs and reduced output, raising unemployment.

⇒ ***The traditional Phillips curve does not hold in the presence of a supply shock ⇔ imitation of an exclusively demand-based approach to understanding inflation.***



III – New Keynesian Phillips Curve

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- Evolution from the Keynesian model to the New Keynesian model: nominal rigidities explain why economic policies can have a short-term effect on the real economy, it is no longer surprises (adaptive $\to$ rational expectations) + microfoundations. Gives rise to the **New-Keynesian Phillips Curve** (NKPC):

$$\pi_t = \beta E_t(\pi_{t+1}) + K(y_t - y^*) \quad (3)$$

$E_t(\pi_{t+1})$ *the rational expectation of future inflation (past inflation, immediately available information, and future beliefs),* K *the slope of the PC and* $y_t - y^*$ *the output gap (real vs. potential output). An alternative version exists with marginal production costs instead of the OG. potentielle).*

$\Rightarrow$ ***Remains a demand-based model, with no integration of supply, so it does not explain the stagflation of the 1970s.***

Extended NKPC : $\pi_t = \beta E_t(\pi_{t+1}) + K(y_t - y^*) + \varepsilon_t \quad (4)$

ε_t *an exogenous shock (supply shock on production costs, fiscal shock, or sector-specific shock).*

- LaThe NKPC fits into DSGE, which is the New Keynesian model (IS / Taylor Rule / NKPC).

IV – Validity and Relevance of the Phillips Curve

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- Current debates focus on the validity and relevance of this relationship in the contemporary economy, particularly in the context of inflation, monetary policy, and global economic conditions.
- Prior to Covid, many economies experienced historically low unemployment rates without facing high inflation.
- Inflation remained relatively stable despite fluctuations in unemployment.

⇒ ***Flattening of the Phillips curve***

- Recent events (Covid crisis and war in Ukraine) show that supply shocks are still poorly captured by the Phillips curve. The pandemic led to supply disruptions, price spikes, and labor shortages in certain sectors, causing inflationary pressures despite relatively high unemployment rates in some regions. The war in Ukraine also led to increases in energy and commodity prices, exacerbating inflationary tensions.

IV – Validity and Relevance of the Phillips Curve

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- **More anchored inflation expectations / more credible central banks (inflation targeting framework):** Anchoring inflation expectations around central bank targets (e.g., 2% inflation) can make the Phillips curve less sensitive to unemployment. When agents expect monetary authorities to act swiftly to return inflation to target, the effect of the unemployment gap on inflation becomes less pronounced.
- **Diversification in supply chains through globalization:** Importing cheaper goods, particularly from low-labor-cost countries, has limited inflationary pressures in advanced economies by reducing their production costs.
- **Technology and innovation:** Technological innovation and digitalization have also had a deflationary effect, by reducing production costs and enabling productivity gains.
- **More structural unemployment:** Increase in the structural component of unemployment due to technological progress (demand for more technical/specialized skills), globalization (loss of industrial jobs in mature economies), demographic shifts (aging population), and stricter regulations (rigid contracts and high firing costs preventing quick labor market adjustment).

PHILLIPS CURVE – CASE STUDY

Case Study

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1) Phillips Curve in the Euro Area – Analysis of the impact by unemployment type

- Compare the Phillips curve in two Euro Area economies, one with predominantly structural unemployment and the other cyclical

2) Phillips Curve in the Euro Area – Analysis of the impact by energy dependence

- Compare the Phillips curve in two Euro Area economies, one heavily dependent on energy imports (especially oil) and the other not

3) Phillips Curve in the Euro Area – Evolution of the slope

- Analyze the Phillips curve over different characteristic sub-periods of the Euro Area (inspired by BdF 2018 paper)

4) Phillips Curve worldwide – Mature VS Emerging/Developing Economies

- Compare the Phillips curve in two global economies, one mature and one emerging (focusing on inflation expectation anchoring)

Case Study

- Report content:
  - 1–3 pages on theory + introduction of the research question + stylized facts + outline
  - 2–5 pages on the econometric application (explanation of methodology and results + tests)
  - 1–2 pages of conclusion (link between theory and econometric results)
- Phillips curve to estimate according to topics and project progress:
  - Inflation ~ unemployment
  - Inflation ~ adaptive expectations + unemployment
  - Inflation ~ adaptive expectations + unemployment + oil (or other energy)
  - Inflation ~ output gap
  - Inflation ~ adaptive expectations + output gap
  - Inflation ~ adaptive expectations + output gap + pétrole (ou autre énergie)

Be careful with the use of generative AI

Statistical Tests (1)

- **Unit root:** When time series are stationary, relationships between variables (e.g., dependency between dependent and independent variables) remain constant over time. This provides reliable parameter estimates. If series are non-stationary, relationships can change over time, making estimates unstable and potentially biased.
  - Augmented Dickey-Fuller test (ADF):
    - H0: Non-stationarity
    - H1: Stationarity

- **No residual autocorrelation:** When residuals are autocorrelated (i.e. dependent on each other), it suggests the model does not capture all information in the data and systematic unexplained patterns remain. This can lead to inefficient coefficient estimates with incorrect standard errors.
  - Durbin-Watson test:
    - H0: No autocorrelation
    - H1: Autocorrelation

Statistical Tests (1)

- **<u>Homoscedasticity of residuals:</u>** Homoscedasticity guarantees that regression coefficient estimates are efficient. If residual variance varies (heteroscedasticity), estimates can be inefficient or standard errors distorted.
  - Breusch-Pagan test:
    - H0: Homoscedasticity
    - H1: Heteroscedasticity

- **<u>Normality of residuals:</u>** Significance tests (t-tests, F-tests) rely on the assumption that residuals follow a normal distribution. If violated, p-values may be misleading, potentially leading to incorrect conclusions on variable significance.
  - Jarque-Bera test
    - H0: Normal distribution
    - H1: Non-normal distribution

EXCHANGE RATES – CONCEPTS

What is the Foreign Exchange Market?

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- **World's largest financial market: > $7.5 trillion USD traded daily**
- **Distinctive financial market**
  - Open 24h/24, 5d/7
  - Decentralized (Over-the-Counter)
  - Immense liquidity, instant transactions
  - Pair quotation
- **Main market participants**
  - Central banks: stabilize currency and/or manage reserves (competitiveness, inflation, volatility)
  - Commercial banks: for their own account or on behalf of clients
  - Multinational corporations: pay foreign suppliers and hedge against exchange rate risk (competitiveness & profitability)
  - Investors & speculators: profit from currency value fluctuations

How the Foreign Exchange Market Works

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Spot Market
Immediate currency exchange

Forward Market
Exchange rate agreed today for a transaction on a specified future date. Ideal for budget planning.

FX Options
Right, but not obligation, to exchange currency at a set rate. Hedging tool against adverse movements.

FX Swaps
Combination of a spot purchase and forward sale. Allows liquidity management without altering risk exposure.

Quotation Principle
The rate expresses the value of the base currency relative to the quote currency. “s” is the exchange rate between A and B.

1**A** = s**B**

Base currency

Quote currency

Exchange Rate Regimes

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- **Fixed exchange rate:** value of a currency is pegged to another currency or basket by monetary authorities (e.g. CFA Franc pegged to Euro).
  - Devaluation vs Revaluation
  - Limits FX risk (+), imports credibility from a strong currency (+), loss of monetary autonomy (-), risk of speculative attacks if FX reserves are low (-)
- **Floating/Flexible exchange rate:** determined entirely by market forces (supply/demand) without government intervention (e.g. EUR/USD).
  - **Depreciation vs Appreciation**
  - Automatic adjustment (+/-), monetary independence (+), external shock risk (-), difficult cost planning if FX rate is volatile (-)
- **Intermediate regimes :** monetary authorities intervene periodically in the foreign exchange market to influence the currency's value without allowing a fully free float (fluctuation bands, managed float, currency basket…). E.g., the PBOC sets a band within which the Yuan can fluctuate.
  - Protection against speculation (+), flexibility to absorb shocks (+), stability for trade (+), lack of transparency (-), high cost in reserves (-)

Quotation Principle

Nominal exchange rate – Screen value:

- It is the price of one currency in terms of another, as displayed on financial markets (e.g., €1 = $1.16).
- Used for immediate transactions, tourism, and capital flows.
- Does not reflect purchasing power. If nominal Euro appreciates by 5% but Eurozone inflation is also 5%, purchasing power remains unchanged.

Real exchange rate – Price competitiveness:

- It is the nominal rate adjusted for the price ratio (inflation) between two countries.
- $Q_t = \frac{s_t * P_t}{P_t^*}$
- If Q_t increases (real appreciation): the country becomes more "expensive", its exports suffer.
- If Q_t decreases (real depreciation): the country gains competitiveness.

Effective exchange rate (EER) – Multilateral view:

- It is a weighted average of exchange rates relative to main trading partners.
- Measures overall strength of a currency in nominal (NEER) and real (REER) terms.
- REER is the key indicator as it combines basket weights and inflation differential effects.
- Example: If Euro appreciates vs Dollar but depreciates vs Yuan (major trade partner), EER may remain stable, indicating overall trade competitiveness is unchanged.

Three Horizons to Understand Exchange Rates

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Short term – Sentiment & Volatility

Dominated by sentiment and market reactions to unexpected shocks

- Risk-on / Risk-off: Flight to safe havens (JPY, CHF, USD) during crises
- Speculative flows: Fund positioning (CFTC data) driving volatility
- Geopolitics: Immediate impact of shocks (conflicts, elections, oil)
- Macroeconomic data surprises

-> *Dornbusch Overshooting Effect*

Medium term – Financial Flows

Driven by yield differentials and relative economic growth prospects

- Interest rate differential: Key driver of carry trade (attraction to higher yields)
- Monetary policy: Anticipation of central bank decisions
- Growth prospects & attractiveness: Direct investment and portfolio flows

-> *Interest Rate Parity and growth differential*

Long term – Fundamentals

Return to fundamental equilibrium values guided by purchasing power parity and trade balances

- Purchasing Power Parity (PPP): Adjustment of exchange rates according to price levels (e.g. Big Mac Index)
- Balance of payments: Structural current account surplus supports currency strength

-> *Purchasing Power Parity and Balassa-Samuelson effect*

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EXCHANGE RATE DETERMINANTS – REVIEW

Purchasing Power Parity (PPP) – Law of One Price and Obstacles

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- **Law of one price:** Under conditions of perfect competition and without trade barriers or transport costs, identical goods should sell at the same price everywhere in the world once exchange rate adjustments are made. The benchmark exchange rate (s_t^{PPP}) is the one that ensures identical purchasing power.

Nominal spot exchange rate:

$$1\ currency = s_t$$

E.g.: 1 EUR = 1.16 USD

Law of one price:

$$P_t * s_t^{PPP} = P_t^*$$

ex.: CPI EUR * EUR/USD = CPI USD

$$s_t^{PPP} = P_t^* / \ P_t$$

Assessment of the spot exchange rate:

$s_t > s_t^{PPP} \ \rightarrow overvaluation$(EUR)

$s_t < s_t^{PPP} \ \rightarrow undervaluation$ (EUR)

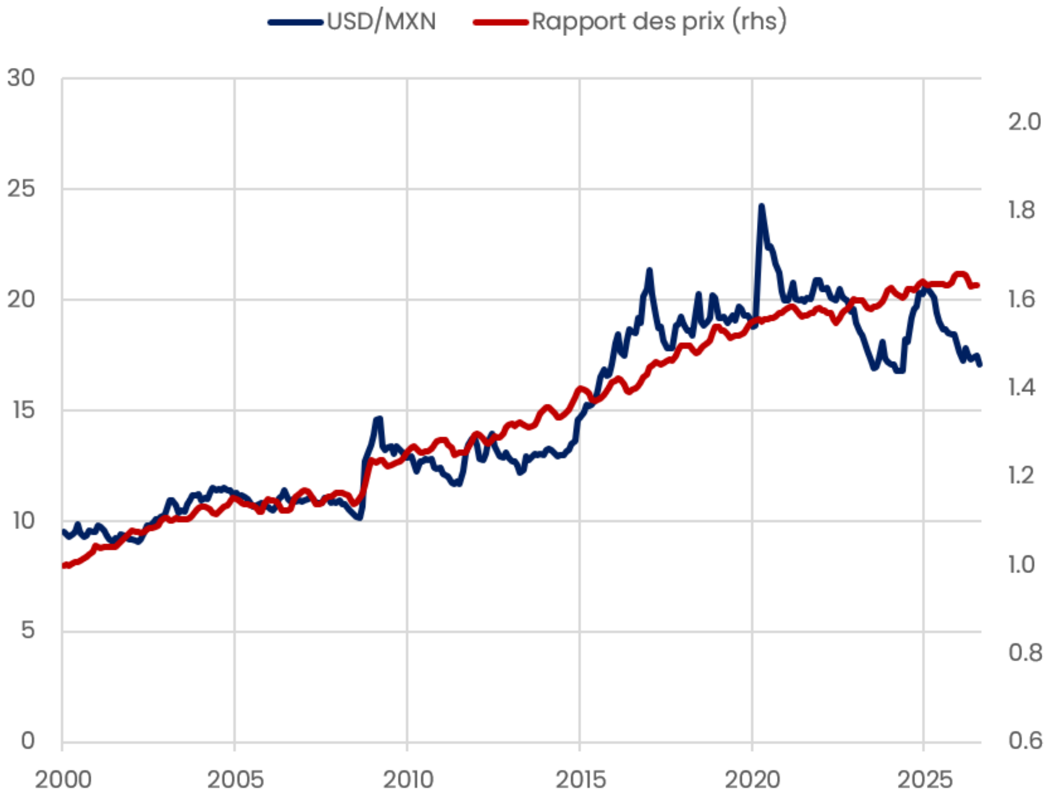
- **PPP does not always hold, especially in the short term:**
  - Trade barriers (tariffs, quotas)
  - Presence of non-tradable goods (local services, labor)
  - Transportation costs and product differentiation
  - Balassa-Samuelson effect

- **Absolute PPP vs. Relative PPP**
  - **Absolute PPP⇔ price ration:** $s_t^{PPP} = P_t^* / \ P_t$
  - **Relative PPP⇔ inflation rate differential (using log-differences):** $\gamma_t^{PPP} = \pi_t^* - \pi_t$ with γ the nominal exchange rate growth rate and π the inflation rate.

Purchasing Power Parity (PPP) –Absolute and Relative

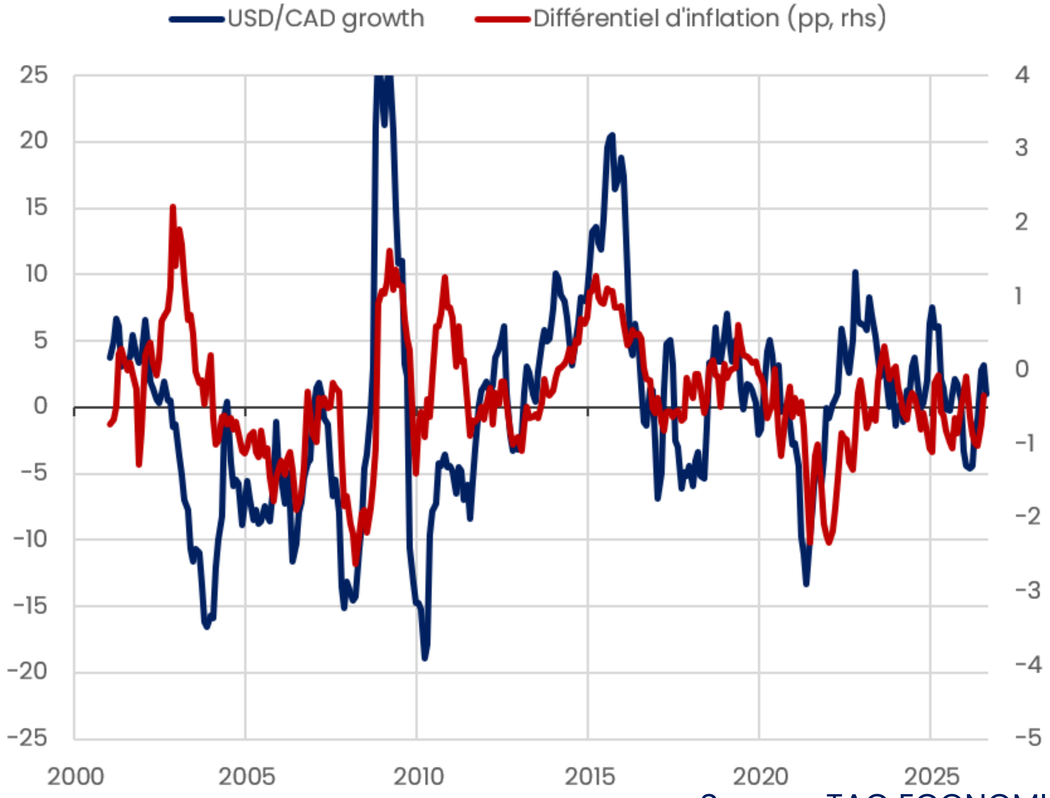
- In practice, relative PPP (inflation rate differential) is more widely used than absolute PPP (price ratio) because it allows for analyzing adjustment dynamics/exchange rate movements rather than simply indicating a theoretical equilibrium level (cf. limitation of absolute PPP).

USD/MXN vs price level ratio



USD/CAD growth vs inflation differential

In % and pp



Source: TAC ECONOMICS Datalab

Interest Rate Parity (IRP)

- **No-arbitrage principle:** Driven by capital mobility, expected returns on two currencies must equalize. If investing 1€ today for a one-month return, arbitrage decision depends on relative returns:

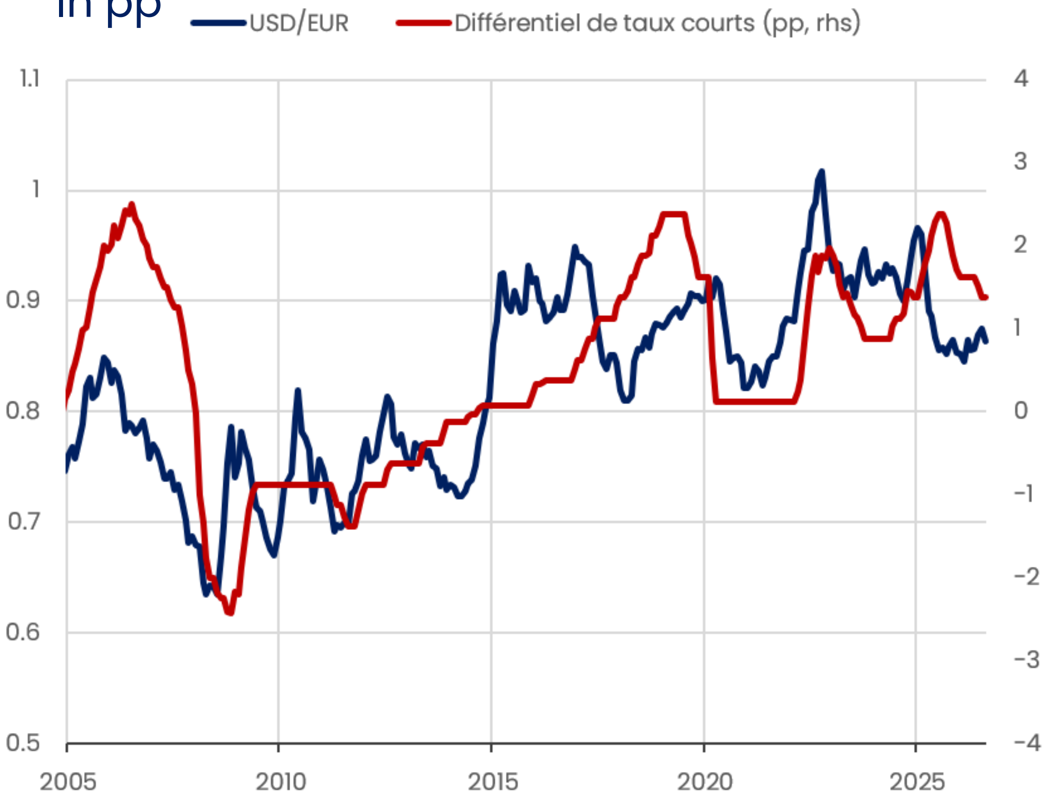
	Decision today (t) 1 EUR = s_t USD	Decision in one month (t+1) 1 EUR = s_{t+1} USD	Conversion to € (t+1) Comparison of returns
EUR Market (€)	1 €	Return on investment (in €) 1 € * $(1 + i_t^{€})$	
USD Market ($)	Conversion en \$ 1 € * s_t	Return on investment (in \$) (1 € * s_t) * $(1 + i_t^{\$})$	Return on investment (in €) $\frac{(1 € * s_t) * (1 + i_t^{\$})}{s_{t+1}}$

- At equilibrium, the returns in red are equivalent, the exchange rate compensates for arbitrage opportunities (interest rate differential).
- In the short run, the exchange rate is not at equilibrium and therefore adjusts:
  - If $i_t^{€} > i_t^{\$}$ => capital inflow into the euro area => ↗ demand for € => EUR appreciation (↗ s_t)
  - If $i_t^{€} < i_t^{\$}$ => capital outflow=> ↗ demand for \$ => EUR depreciation (↘ s_t)

Interest Rate Parity (IRP)

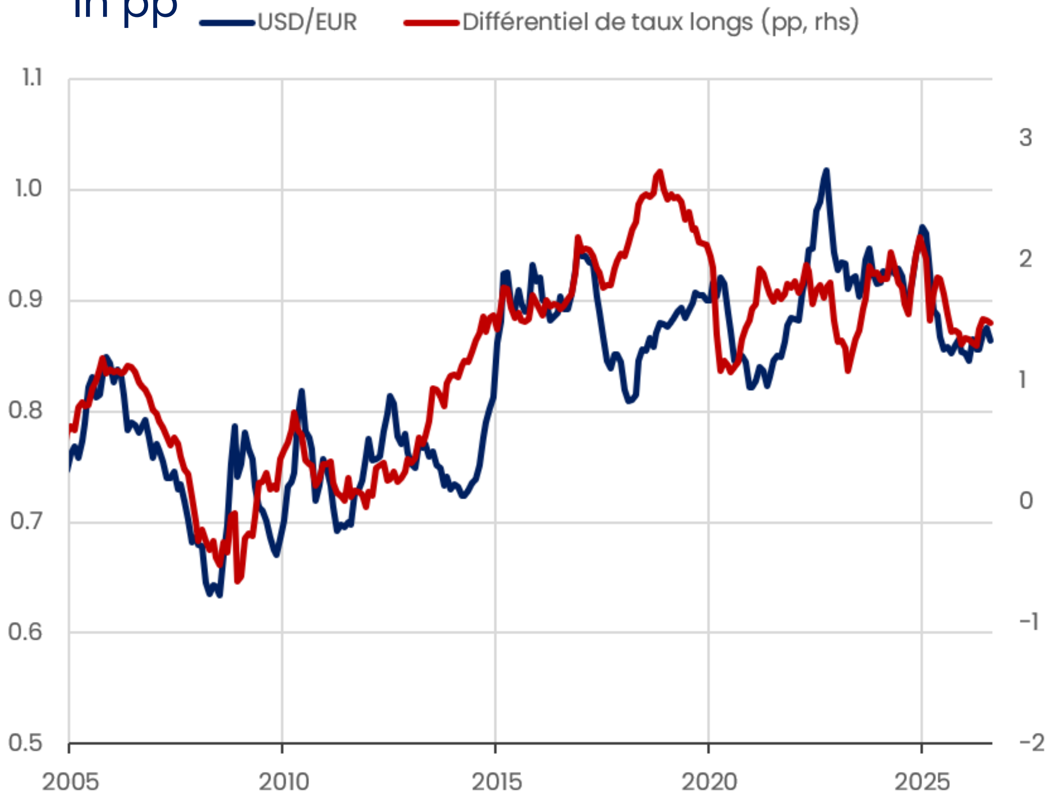
USD/EUR vs short-term rate differential

In pp



USD/EUR vs long-term rate differential

In pp



- The short-term rate differential (IRP) is imperfect for explaining exchange rate dynamics as it remains a short-term factor with higher inertia than FX markets (central bank meetings held at most once a month). For mature economies, the long-term rate differential (10-year bonds) tracks FX dynamics better by incorporating IRP (expectations theory of interest rates) alongside other key market factors (debt, growth, inflation, and risk perception).

PPA & PTI – CASE STUDY

Case Study

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1) **PPP: Absolute vs Relative**
   - Calculate/estimate both PPP versions (absolute/relative) for **at least 2 economies**
2) **IRP: Policy Rate vs Long-Term Rate**
   - Calculate/estimate both IRP versions (short rate/long rate) for **at least 2 economies**
3) **PPP & IRP: Mature vs Emerging Economies**
   - Calculate/estimate PPP and IRP for **at least 2 economies (1 mature, 1 emerging)** + combined PPP/IRP in one estimation
4) **PPP & IRP: Fundamental and Complementary Determinants**
   - Estimate a combined PPP/IRP version and an augmented combined PPP/IRP version with additional determinants for **at least 3 economies**

Case Study

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  - 2–5 pages on the econometric application (explanation of methodology and results + tests)
  - 1–2 pages of conclusion (link between theory and econometric results)

- For certain subjects, PPP and IRP must first be calculated without an econometric model :
  - **Absolute PPP (non-estimated):** Convert both CPIs to base 100 in the same year, then scale by the exchange rate
  - **Relative PPP (non-estimated):** Simple inflation rate difference, then scale by the exchange rate (growth rate)
  - **IRP (non-estimated):** Simple interest rate difference, then scale by the exchange rate (growth rate)
  - **Absolute PPP (estimated):** $s_t = \beta_0 + \beta_1 \times \frac{P_t^*}{P_t} + \varepsilon_t$ (in log by convention)
  - **Relative PPP (estimated):** $s_t = \beta_0 + \beta_1 \times (\pi_t^* - \pi_t) + \varepsilon_t$ et/ou $s_t = \beta_0 + \beta_1 \times \pi_t^* + \beta_2 \times \pi_t + \varepsilon_t$
  - **IRP (estimated):** $s_t = \beta_0 + \beta_1 \times (i_t^* - i_t) + \varepsilon_t$ et/ou $s_t = \beta_0 + \beta_1 \times i_t^* + \beta_2 \times i_t + \varepsilon_t$ (i_t policy rate then long-term rate for topic 2)
  - **Combined PPP & IRP (estimated):** $s_t = \beta_0 + \beta_1 \times (i_t^* - i_t) + \beta_2 \times (\pi_t^* - \pi_t) + \varepsilon_t$ and/or $s_t = \beta_0 + \beta_1 \times i_t^* + \beta_2 \times i_t + \beta_3 \times \pi_t^* + \beta_4 \times \pi_t + \varepsilon_t$
  - **Augmented combined PPP & IRP (estimated):** $s_t = \beta_0 + \beta_1 \times (i_t^* - i_t) + \beta_2 \times (\pi_t^* - \pi_t) + \beta_3 \times X3_t + [\ldots] + \varepsilon_t$ and/or

    $$s_t = \beta_0 + \beta_1 \times i_t^* + \beta_2 \times i_t + \beta_3 \times \pi_t^* + \beta_4 \times \pi_t + \beta_5 \times X5_t + [\ldots] + \varepsilon_t$$
  - s_t the level or growth rate of the exchange rate depending on stationarity tests and the version of the theory being tested.

Statistical Tests (1)

- **Unit root:** When time series are stationary, relationships between variables (e.g., dependency between dependent and independent variables) remain constant over time. This provides reliable parameter estimates. If series are non-stationary, relationships can change over time, making estimates unstable and potentially biased.
  - Augmented Dickey-Fuller test (ADF):
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- **No residual autocorrelation:** When residuals are autocorrelated (i.e. dependent on each other), it suggests the model does not capture all information in the data and systematic unexplained patterns remain. This can lead to inefficient coefficient estimates with incorrect standard errors.
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  - Jarque-Bera test
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Contacts

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